INTRODUCTION TO SOFTWARE ENGINEERING

UNIT -I

What is software?

- It is any set of machine-readable instructions that directs a computer's **processor** to perform specific operations

- It is inclusive of both **machine instructions** (the **binary code** that the processor "understands") and **source code** (more human-understandable instructions that must be rendered into machine code by **compilers** or **interpreters** before being executed).

- software is a set of programs, procedures, functions, associated data and/or its documentation

Example

- Build a House

Plan

Draw the Design

Buy materials

Build

Check if everything works

Give the keys to owner

- WhatsApp

Your chat database

Install procedure

Terms of Service Document

Server side code which you never see

s/w is not just code. It’s whole package

1.Set of programs 2.Data 3. Documentation(user manuals) 4.procedure(installation)

What is S/W engineering

- It covers not only the technical aspects of building software systems, but also managing issues, such as directing programming teams, scheduling, and budgeting etc.

- It is the application of a systematic, disciplined, quantifiable approach to the design, development, operation, and maintenance of **software**, and the study of these approaches; that is, the application of **engineering** to software.

- In general it is the act of using insights to conceive, model and scale a solution to a problem

THE EVOLVING ROLE OF S/W

It performs dual role

As a product

- Which resides in mobile phone
- Which operates inside a computer
- Which transforms, produce, manage, acquires, modifies, displays information

A vehicle to deliver a product

- Acts as a basis for the control of computer
- Communicates the information
- Creation and control of other programs(s/w tools and environment)

The programmer of an earlier era has been replaced by the teams of s/w specialists, each focusing on one part of the technology required to deliver a complex application . And yet the questions rises as follows

- Why does it take so long to get software finished?
- Why are developers costs so high?
- Why can't we find all the errors before we give the software to customers?
- Why do we continue to have difficulty in measuring the progress as software is being developed?

These questions demonstrate the industry's concern about software and the manner in which it is developed ,And that concern lead to the adoption of s/w engineering practices

Software

A text book definition of software might take the following form

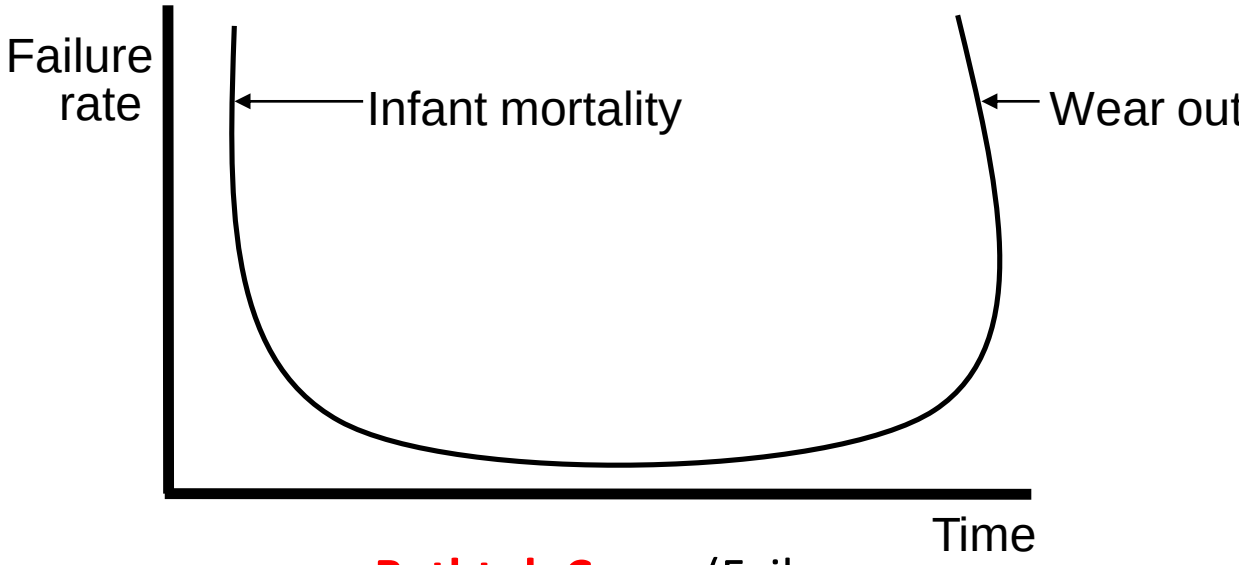
- Instruction
- Data structures
- Documents

- characteristics that are considerably different than those of H/w
- S/w is developed or engineered, it's not manufactured in the classical sense
- S/w doesn’t “wear out”
- Although the industry is moving toward component-based construction , most s/w continues to be customer built

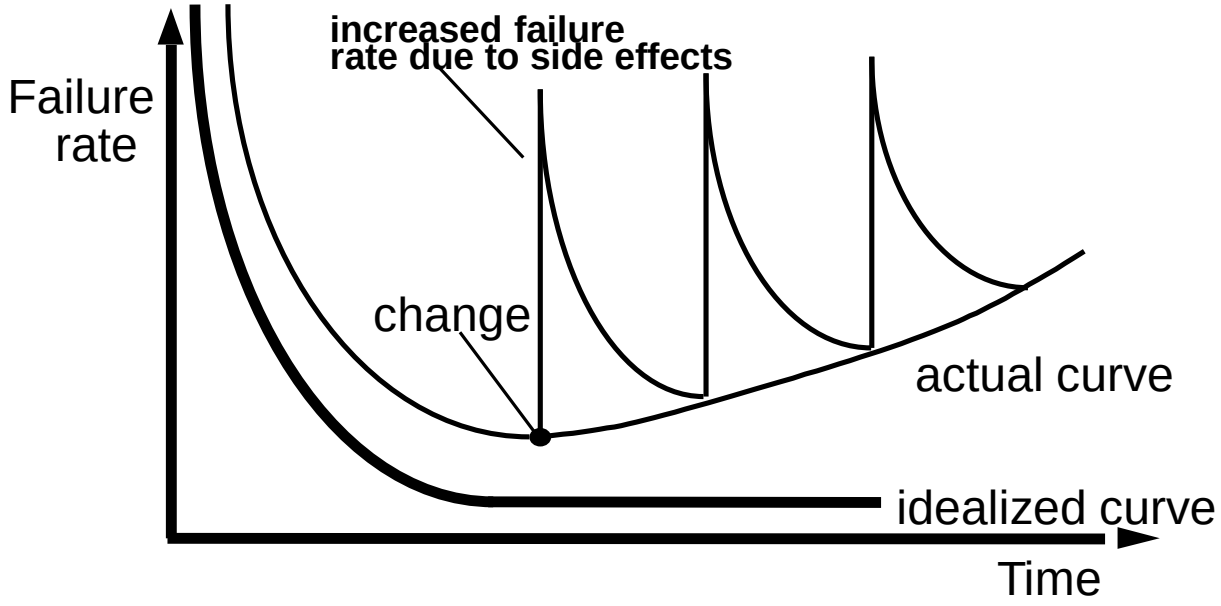
1. S/w is developed or engineered, its not manufactured in the classical sense

- The manufacturing phase for h/w can introduce quality problems which doesn't exists for s/w
- The relationship between people applied and the work accomplished is entirely different.
- Both activities require the construction of a product but the approaches are different

2. S/w doesn't "wear out"



Bathtub Curve (Failure curve for hardware)



Failure curves for software

3. Although the industry is moving toward component-based construction , most s/w continues to be customer built

➢Hardware for the computer is designed and built.

➢But for the software, component should be designed and implemented so that it can be reused in many different programs.

The changing nature of s/w

System software

- It's a collection of programs written to serve other programs
- Eg : Windows,android,compilers , editor, drivers etc.

Application software

- It consists of standalone programs that solve a specific business need.

Engineering/scientific software

- Engineering and scientific software application range from astronomy to volcanology from automatic stress analysis.

Embedded s/w

- It resides within a product or system and control features and functions for the end user.

Product line s/w

- Designed to provide a specific capability that can be used by many customers
- e.g., spread sheets, graphics, DBM etc.

Web applications

- It can be little more than set of linked hypertext that present information using text and limited graphics.

Artificial intelligence s/w

- This makes use of non numerical algorithms to solve complex problems.

Software myths

Myths are classified as

- **Management myths**
- **Customer myths**
- **Practitioner's myths**

Management myths

- **Myth1:** We already have a book that's full of standards and procedures for building software, won't that provide my people with everything they need to know.
- **Myth2:** If we get behind schedule, we can add more programmers and catch up.
- **Myth3**: If I decide to outsource the software project to a third party, I can just relax and let that firm build it

Customer myths

- **Myth1**: A General statement of objectives is sufficient to begin writing programs – we can fill in the details later.
- **Myth2**: Project requirements continually change, but change can be easily accommodated because software is flexible.

Practitioner's myths

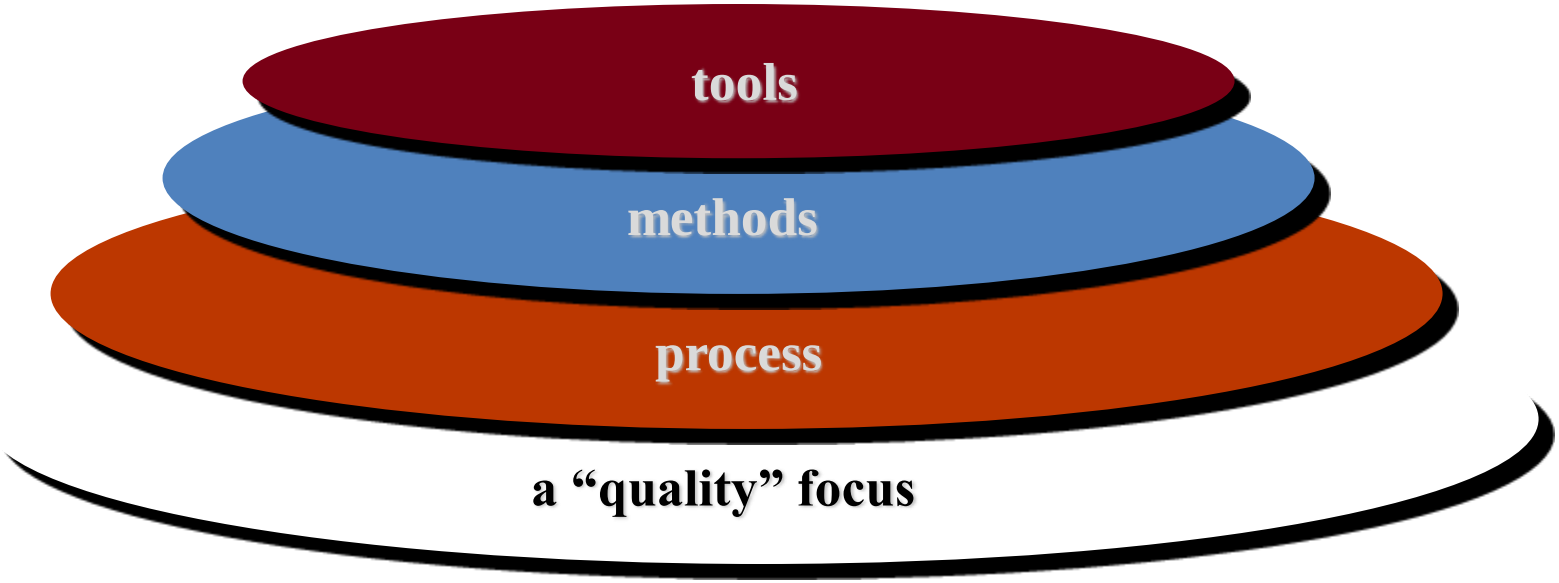
- **Myth1**: once we write the program and get it work, our job is done.
- **Myth2**: until I get the program running I have no way of assessing its quality.
- **Myth3**:The only deliverable work product for a successful project is the working program.
- **Myth4**:Software engineering will make us create voluminous and unnecessary documentation and will invariably slow us down.

A Generic view of process

S/w engineering-A layered technology

- Definition
  - Software Engineering is “the establishment and use of sound engineering principles in order to obtain economically software that is reliable and work efficiently on real machines”.
- IEEE
  - The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software.
  - The study of approaches described above.

S/w engineering layers



A quality focus layer

- Any engineering approach must rest on an organizational commitment to quality(reliability, efficiency etc)
- This culture leads to the development of more effective approaches to s/w engineering.
- The bedrock that supports s/w engineering is quality focus.

Process layer

- The foundation for s/w engineering is process layer
- It allows development of s/w on time
- Work products are produced, and milestones are established.
- It forms the base for management control on software product

Method

- It includes a broad array of tasks that includes communication, requirement analysis , design , construction, testing and support.
- s/w engineering methods provides knowledge for building a s/w .

Tools

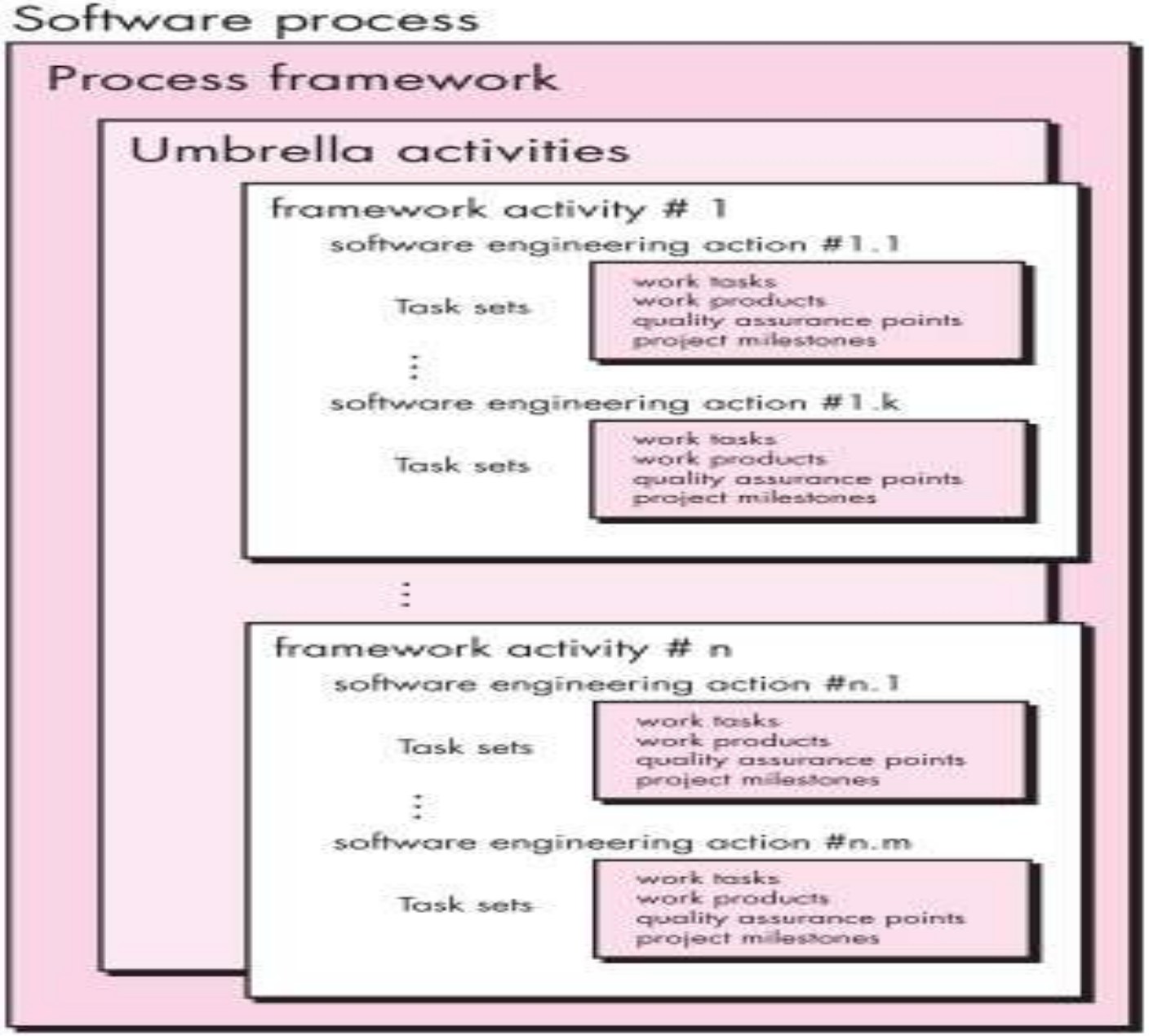
- The tools layer provide automated or semi automated support for the process and method layers..
- The information created by one tool can be used by another
- A system for the support of s/w development called CASE(computer-aided s/w engineering) is established.
- CASE help in application development including analysis, design, code generation, testing debugging etc.

Process Models

A Generic process Model

A Process Framework

- A process framework establishes the foundation for a complete s/w process.
- The process framework contains framework activities that are applicable to all s/w projects.
- The process framework also contains a set of umbrella activities that are applicable for the entire s/w process.


Software process
Process framework
Umbrella activities
framework activity # 1
software engineering action #1.1
Task sets
work tasks
work products
quality assurance points
project milestones
software engineering action #1.k
Task sets
work tasks
work products
quality assurance points
project milestones
framework activity # n
software engineering action #n.1
Task sets
work tasks
work products
quality assurance points
project milestones
software engineering action #n.m
Task sets
work tasks
work products
quality assurance points
project milestones

- Each frame work activity is populated by set of s/w engineering actions

- Each action is populated with individual work tasks which accomplish some part of work implied by the action.

The following generic process framework which is applicable to majority of s/w project.(5 framework activities)

Communication

- Collaboration with customer
- Gathering requirements

Planning

- Technical task to be conducted
- The risks that may occur.
- The resources that require
- The work products and work schedules

Modeling

- It allows the developer and the customer to understand requirements and the design to achieve those requirements.

Construction

Coding

- Testing

Deployment

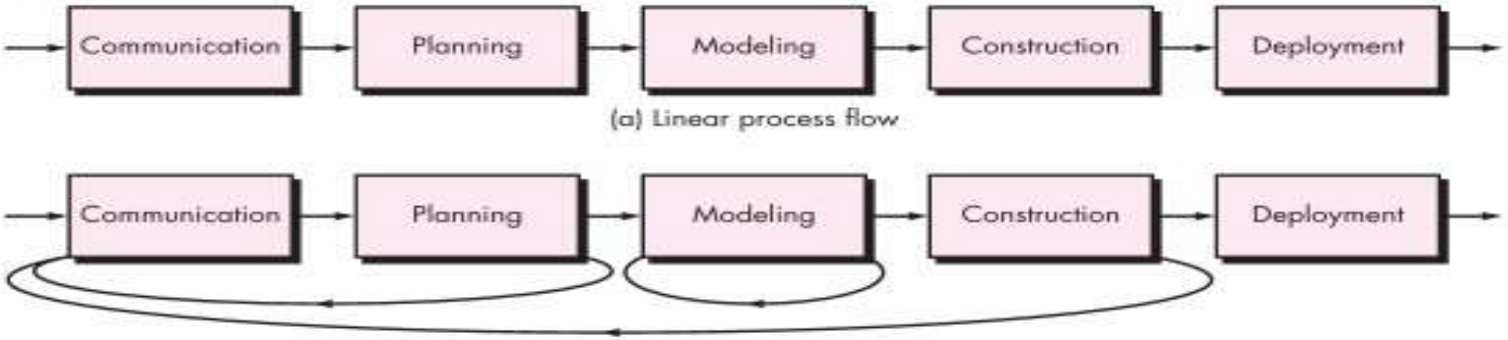
- Delivery of the product to the customer.
- Feedback by the customer.

The framework will be composed of umbrella activities

- S/w project tracking and control
- Risk management
- S/w quality assurance
- Formal technical reviews
- Measurement
- s/w configuration management
- Reusability management
- Work product preparation and production

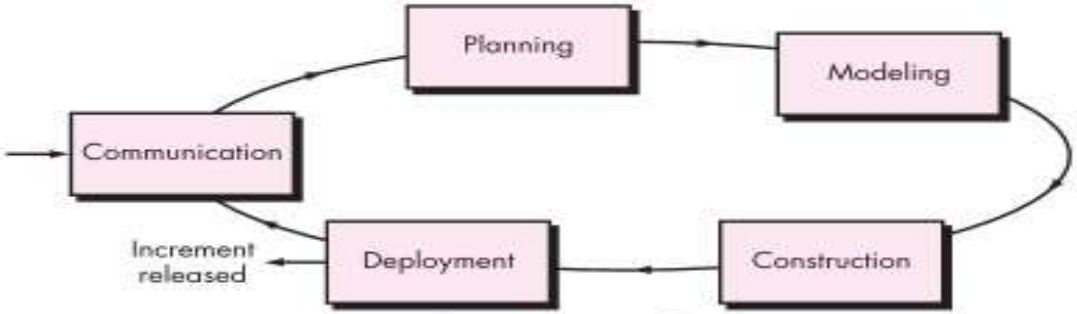
All process model can be characterized with in the process frame work, but process model do differ time to time

FIGURE 2.2 Process flow

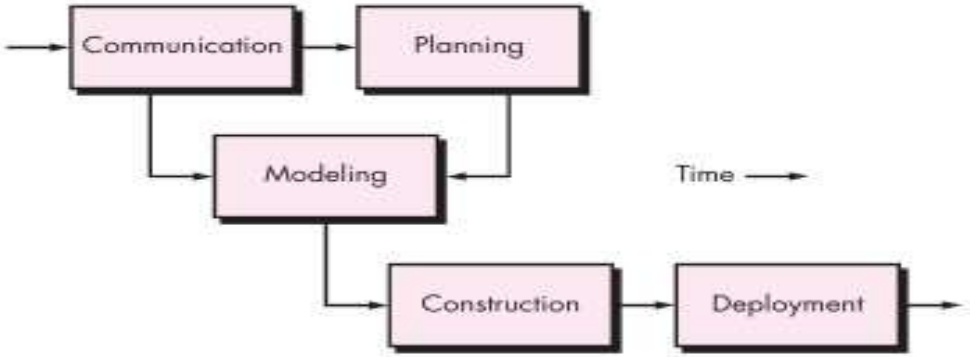


(a) Linear process flow

(b) Iterative process flow



(c) Evolutionary process flow



(d) Parallel process flow

Process Assessment and Improvement

CMMI

The software engineering institute uses an assessment that results in a five point grading scheme for process maturity in compliance with capability maturity model.

The CMMI represents a process models in 2 different ways

- Continuous model.
- Staged model.

Each process area is assessed against specific goals and practices and is rated according to the following capability levels.

Represents in two ways

1.Continuous model: 6 levels to achieve

- Level 0 (Incomplete) :process area not performed/not achieve all goals
- Level 1 (Performed) :specific goals of process area defined by CMMI should be achieved.
- Level 2 (Managed) :level 1 criteria should be achieved & work tasks , work products should be monitored , controlled & reviewed.
- Level 3 (Defined) :level 2 criteria should be achieved & improve information according to organizational guide lines.
- Level 4 (Quantitatively managed) :level 3 criteria should be achieved & process area should be controlled & improved using measurements and quality assessments
- Level 5 (Optimized) :level 4 criteria should be achieved & should meet changing customer needs and continuously improve the efficiency of process area

Some of the specific goals defined for project planning

- **SG1** Establishing estimates:
  - **SP1** Scope of the project
  - **SP2** Work product &task attributes
  - **SP3** Define project life cycle
  - **SP4** Effort & cost

- **SG2** Develop a project plan:
  - **SP1** Budget & scheduling
  - **SP2** Project risk
  - **SP3** Plan for data management
  - **SP4** Plan for project resources
  - **SP5** Plan for needed knowledge & skills
  - **SP6** Stake holder involvement
  - **SP7** Project plan

- **SG3** Commitment to plan
  - **SP1** Review plans
  - **SP2** Reconcile work & resources
  - **SP3** Obtain plan commitment

Some of the generic goals

- **GGI** Achieve specific goals
  - **GP1** perform base practices

- **GG2** Institutionalize a managed process
  - **GP1** organizational policy
  - **GP2** plan the process
  - **GP3** provide resources
  - **GP4** assign responsibilities
  - **GP5** train people
  - **GP6** manage configuration
  - **GP7** involve relative stakeholders
  - **GP8** monitor & control the process
  - **GP9** objectively evaluate adherence
  - **GP10** review status with high level management

- **GG3** Institutionalize a defined process
  - **GP1** establish define process
  - **GP2** collect improvement information

- GG4 Institutionalize a quantitative managed process
- GP1 quantitative objectives for the process
- GP2 stabilize sub process performance

- GG5 Institutionalize an optimizing process
- GP1 ensure continuous process improvement
- GP2 correct root cause of problem

- **Staged CMMI Mode**
- Primary difference is that the staged model defines 5 maturity levels & continuous model defines 6 capability levels
- To achieve maturity levels SG & SP associated with a set of process area must be achieved
- Process areas required to achieve maturity level are give in the table below

Level	Focus	Process area
Optimizing	Continuous process improvement	organizational innovation &development analysis & resolution
Quantitatively managed	Quantitative management	Organizational process performance ,quantitative project management
Defined	Process standardization	Requirements development, technical solutions, verification ,validation ,etc
Managed	Basic project management	Requirements management , project planning etc….
Performed		

Process Models

Process Models

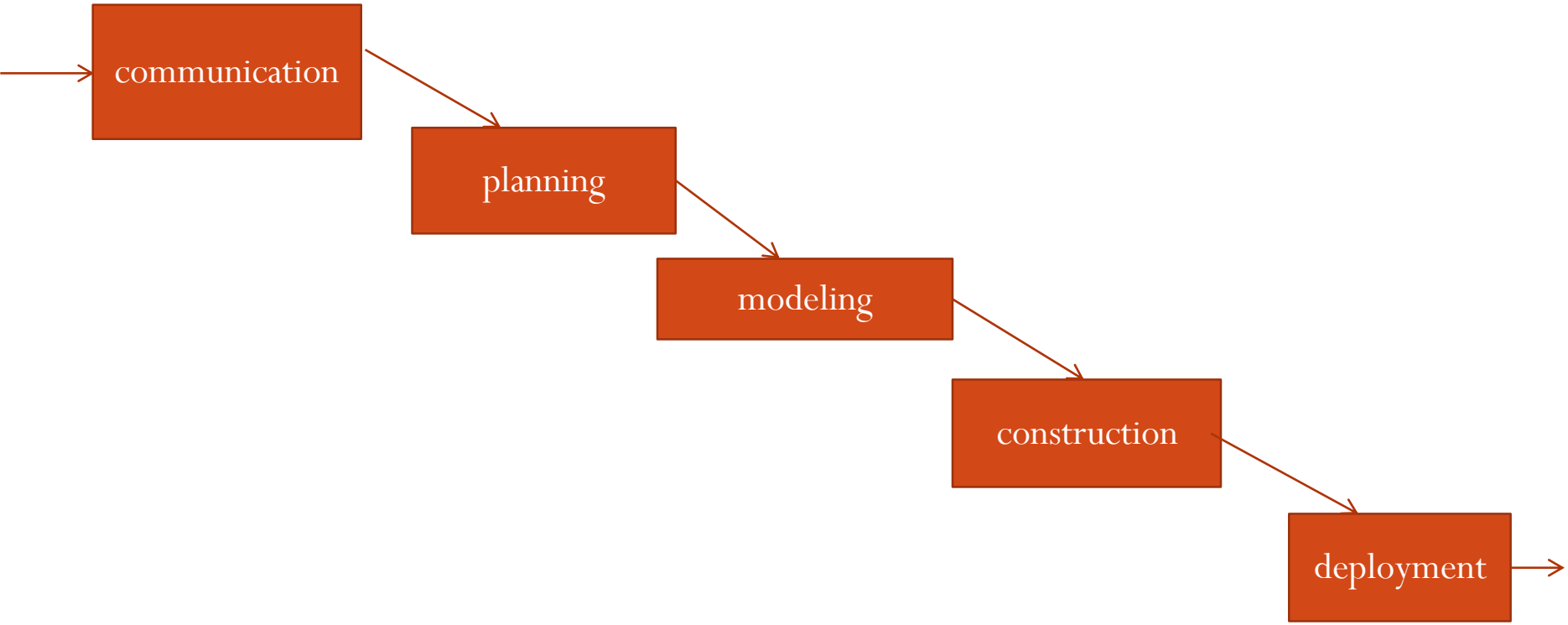
- Every s/w organization should describe a unique set of frame work activities such as communication, planning , modeling, construction ,and deployment.
- Each process model describes a workflow in which the process elements are interrelated to one other.

These are the types of process models

- Waterfall model
- Incremental model
- RAD model
- Prototyping
- Spiral
- Concurrent development model
- Unified process model

The Waterfall Model

- Water fall model is also called as classic life cycle model
- This is oldest paradigm for s/w engineering.



- **Communication:**

Project initiation, requirements gathering .

- **Planning:**

Estimation, scheduling , tracking

- **Modeling:**

Analysis & design

- **Construction:**

Coding and testing

- **Deployment:**

Delivery, support , feedback

When to use the waterfall model:

- This model is used only when the requirements are very well known, clear and fixed.
- Product definition is stable.
- Technology is understood.
- There are no ambiguous requirements
- The project is short.

Advantages of waterfall model:

- This model is simple and easy to understand and use.
- It is easy to manage due to the rigidity of the model
- In this model phases are processed and completed one at a time. Phases do not overlap.

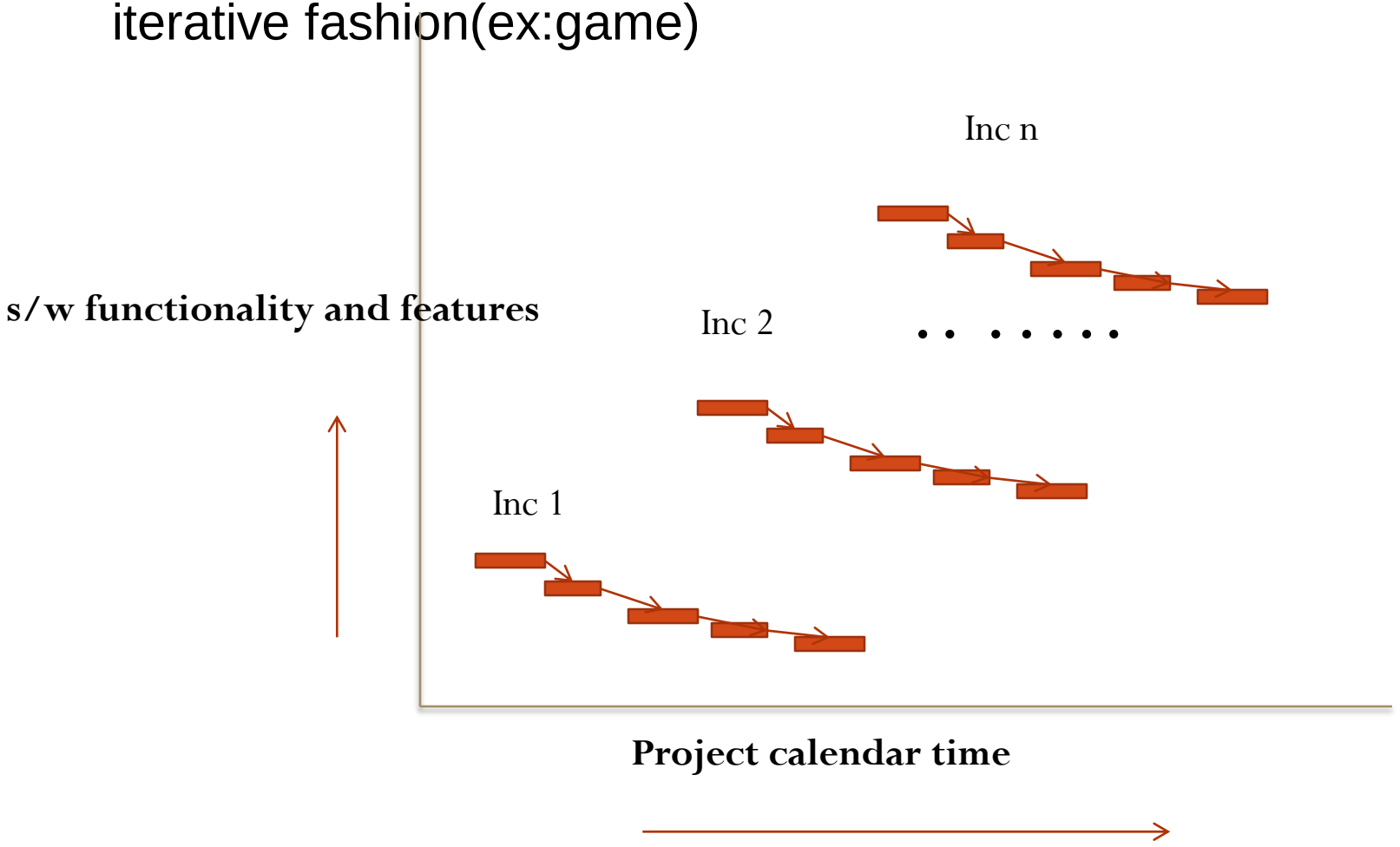
Disadvantages of waterfall model:

- Once the process is done, it is very difficult to go back and change something
- No working software is produced until late during the life cycle.
- High amounts of risk and uncertainty.
- Not a good model for complex and object-oriented projects.
- Poor model for long and ongoing projects.
- Not suitable for the projects where requirements are at a moderate to high risk of changing.

Incremental process model

Incremental model

This model combines elements of waterfall model applied in iterative fashion(ex:game)



- The incremental model combines elements of the linear sequential model with the iterative philosophy of prototyping.
- The incremental model applies linear sequences in a staged fashion.
- Each linear sequence produces a deliverable “increment” of the software.
- The **first increment is often a core product**.
- The **core product is evaluated, a plan is developed for the next increment.**
- The plan addresses the modification of the core product to better meet the needs of the customer and delivery of supplementary features.
- The **process is repeated until the complete product is produced.**

When to use the Incremental model:

- This model can be used when the requirements of the system are **clearly defined and understood**.
- **Major requirements** must be defined; however, some details can evolve with time.
- There is a **need to get a product to the market early**.
- Resources with **needed skill set** are **not available**.

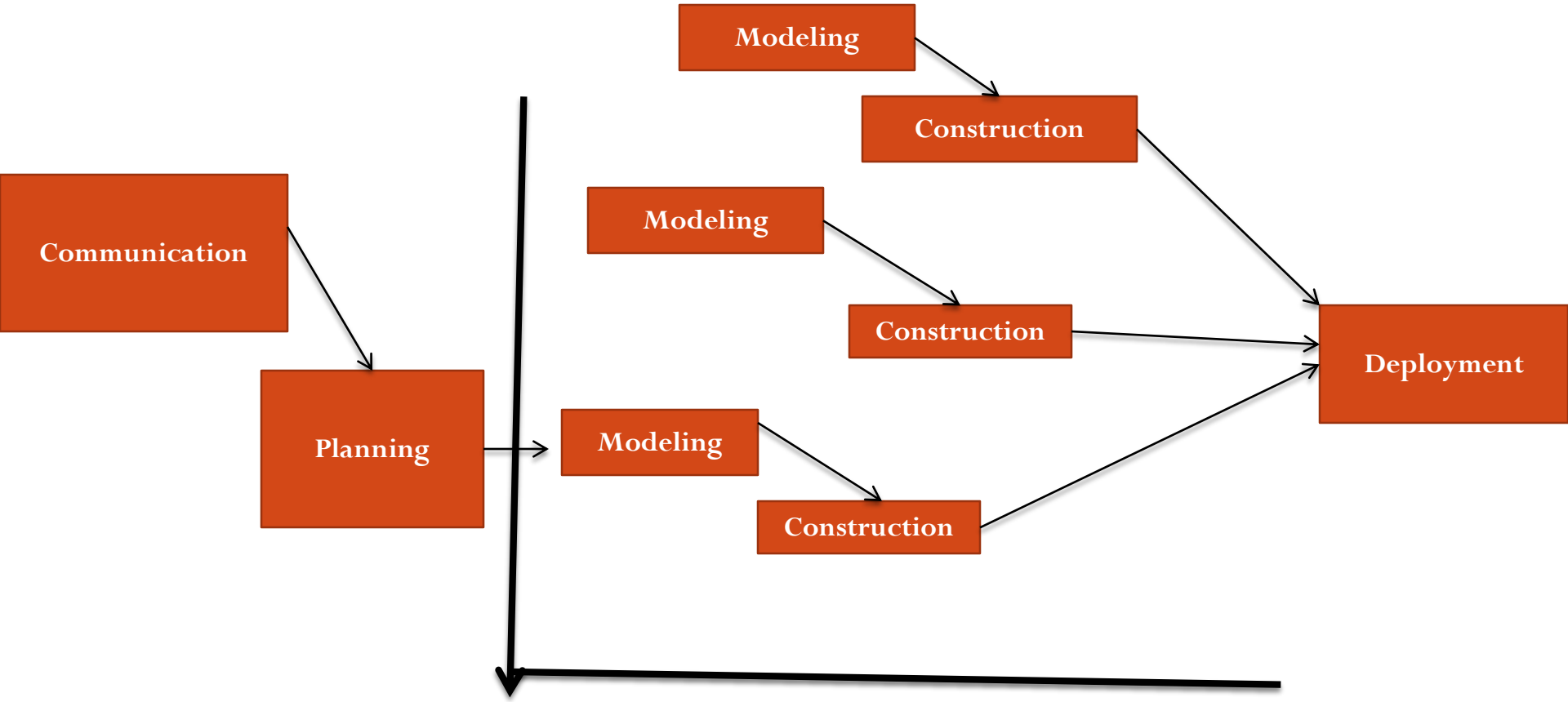
Advantages of Incremental model:

- Generates working software quickly .
- This model is more flexible – less costly to change scope and requirements.
- It is easier to test and debug during a smaller iteration.
- In this model customer can respond to each built.
- Easier to manage risk because risky pieces are identified and handled during it's iteration.

Disadvantages of Incremental model:

- Needs good planning and design.
- Needs a clear and complete definition of the whole system before it can be broken down and built incrementally.
- Total cost is higher than **waterfall.**

The RAD Model



Rapid application development model is a high speed adaptation. in which rapid development is achieved by using a **component based construction approach**.(reusable code)

This process enables the developer to create a fully functional system" with in a very short time(e.g., 60 to 90 days)

- **Communication** is required to understand the business problems.
- **Planning** is essential because multiple s/w teams work in parallel in different system functions.
- **Modeling** has 3 phases -**business, data and process** modeling which gives the design representation that serves as basic for RAD's Construction phase
- **Construction** is all about the **application of code generation**.
- **Deployment** was done for subsequent iterations if necessary.

When to use RAD model:

- RAD should be used when there is a need to create a system that can be modularized in 2-3 months of time.
- It should be used if there's high availability of designers for modeling and the budget is high enough to afford their cost.
- RAD should be chosen only if resources with high business knowledge are available and there is a need to produce the system in a short span of time (2-3 months).

Advantages of the RAD model:

- Reduced development time.
- Increases reusability of components
- Quick initial reviews occur

Disadvantages of RAD model:

- Only system that can be modularized can be built using RAD
- Requires highly skilled developers/designers.
- High dependency on modeling skills
- Inapplicable to **cheaper projects** as cost of modeling and automated code generation is very high.

Evolutionary Process Models

These models are iterative, they are characterized in a manner that enables software engineers to develop increasingly more complete version of the software.

- Prototyping
- Spiral model
- The concurrent development model

Prototyping

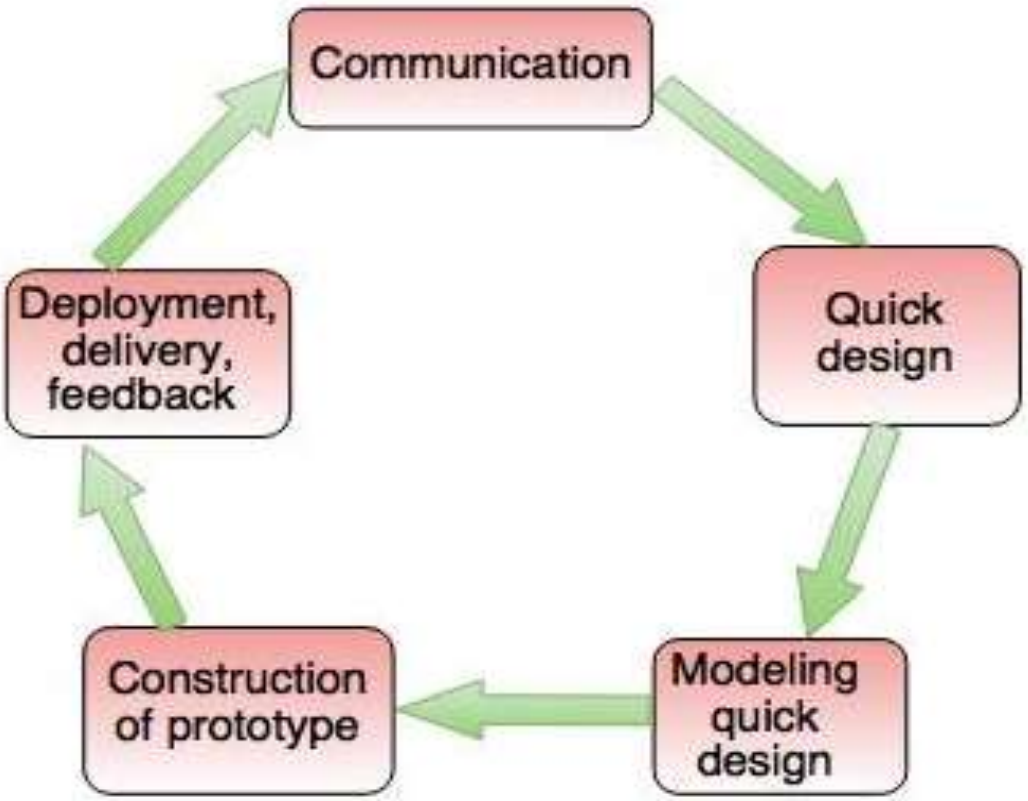


Fig. - The Prototyping Model

Communication

- In this phase, developer and customer meet and discuss the overall objectives of the software.

Quick design

- Quick design is implemented when requirements are known.
- It includes only the important aspects like input and output format of the software.
- It focuses on those aspects which are visible to the user rather than the detailed plan.
- It helps to construct a prototype.

Modeling quick design

- This phase gives the clear idea about the development of software because the software is now built.
- It allows the developer to better understand the exact requirements.

Construction of prototype

The prototype is evaluated by the customer itself.

Deployment, delivery, feedback

- If the user is not satisfied with current prototype then it refines according to the requirements of the user.
- The process of refining the prototype is repeated until all the requirements of users are met.
- When the users are satisfied with the developed prototype then the system is developed on the basis of final prototype.

When to use Prototype model

- Prototype model should be used when the desired system needs to have a **lot of interaction with the end users**.
- Typically, **online systems, web interfaces** have a very **high amount of interaction with end users**, are best suited for Prototype model. It might take a while for a system to be built that allows ease of use and needs minimal training for the end user.
- Prototyping ensures that the end users constantly work with the system and provide a feedback which is incorporated in the prototype to result in a useable system. They are **excellent for designing good human computer interface systems**.

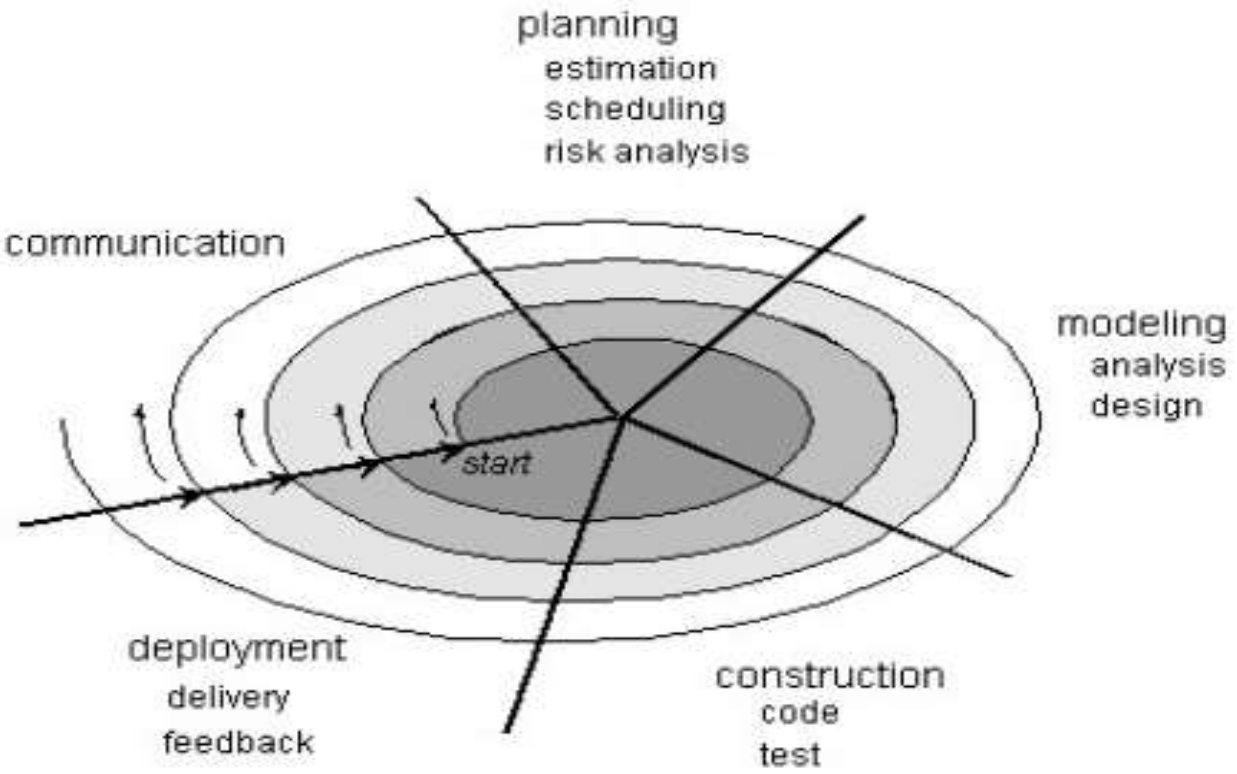
Advantages of Prototyping Model

- Prototype model **need not know** the **detailed input, output**, processes, adaptability of operating system and full machine interaction.
- In the development process of this model **users** are actively **involved**.
- The development process is the best platform to understand the system by the user.
- **Errors** are detected much earlier.
- Gives **quick user feedback** for better solutions.
- It **identifies the missing functionality easily**. It also **identifies the confusing or difficult functions**.

Disadvantages of Prototyping Model :

- The **client involvement is more** and it is not always considered by the developer.
- It is a **slow process** because it takes more time for development.
- **Many changes can disturb** the rhythm of the development team.
- **Users are confused** with the prototype given.

The Spiral model



When to use Spiral model

- When costs and risk evaluation is important
- For medium to high-risk projects (banking,medical,safety related s/w)
- Users are not sure of their needs
- Requirements are complex
- New product line
- Significant changes are expected

Advantages of Spiral model

- High amount of risk analysis hence, avoidance of Risk is enhanced.
- Good for large and mission-critical projects.
- Additional Functionality can be added at a later date.
- Software is produced early in the software life cycle.

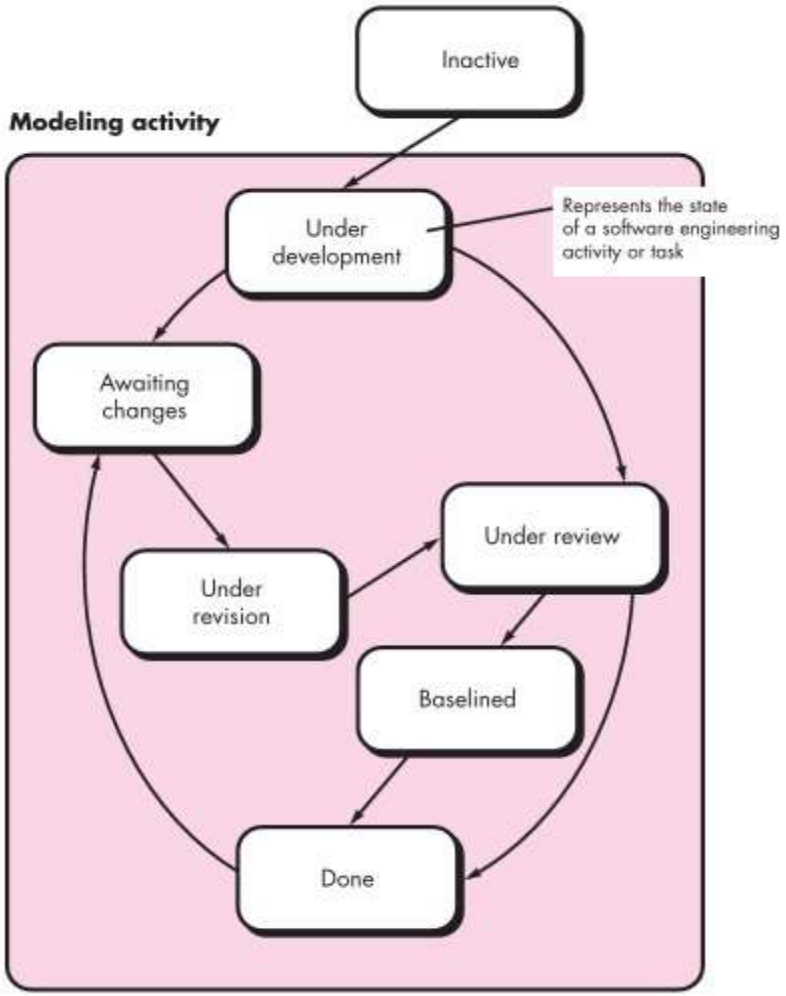
Disadvantages of Spiral model

- Can be a costly model to use.
- Risk analysis requires highly specific expertise.
- Project's success is highly dependent on the risk analysis phase.
- Doesn't work well for smaller projects.

The Concurrent development model

FIGURE 2.8

One element of the concurrent process model



- The concurrent development model, some times called concurrent engineering.

- The concurrent process model can be represented as a series of major technical activities, tasks, and their associated states.

- Each concurrent activity can be in any one of the following states:
  - None
  - Awaiting
  - Under development
  - Done
- The concurrent process model is applicable to all types of s/w development and provide an accurate picture of the current state of a project.

When to use concurrent development model

- If stakeholder's want to know the current state of the project.
- If we want to complete the project in less time.
- If we have more human resources.
- Overall requirements are not needed to start the project.

Advantages of the concurrent development model

- This model is applicable to all types of software development processes.
- It is easy for understanding and use.
- It gives immediate feedback from testing.
- It provides an accurate picture of the current state of a project.

Disadvantages of the concurrent development model

- It needs better communication between the team members. This may not be achieved all the time.
- It requires to remember the status of the different activities.

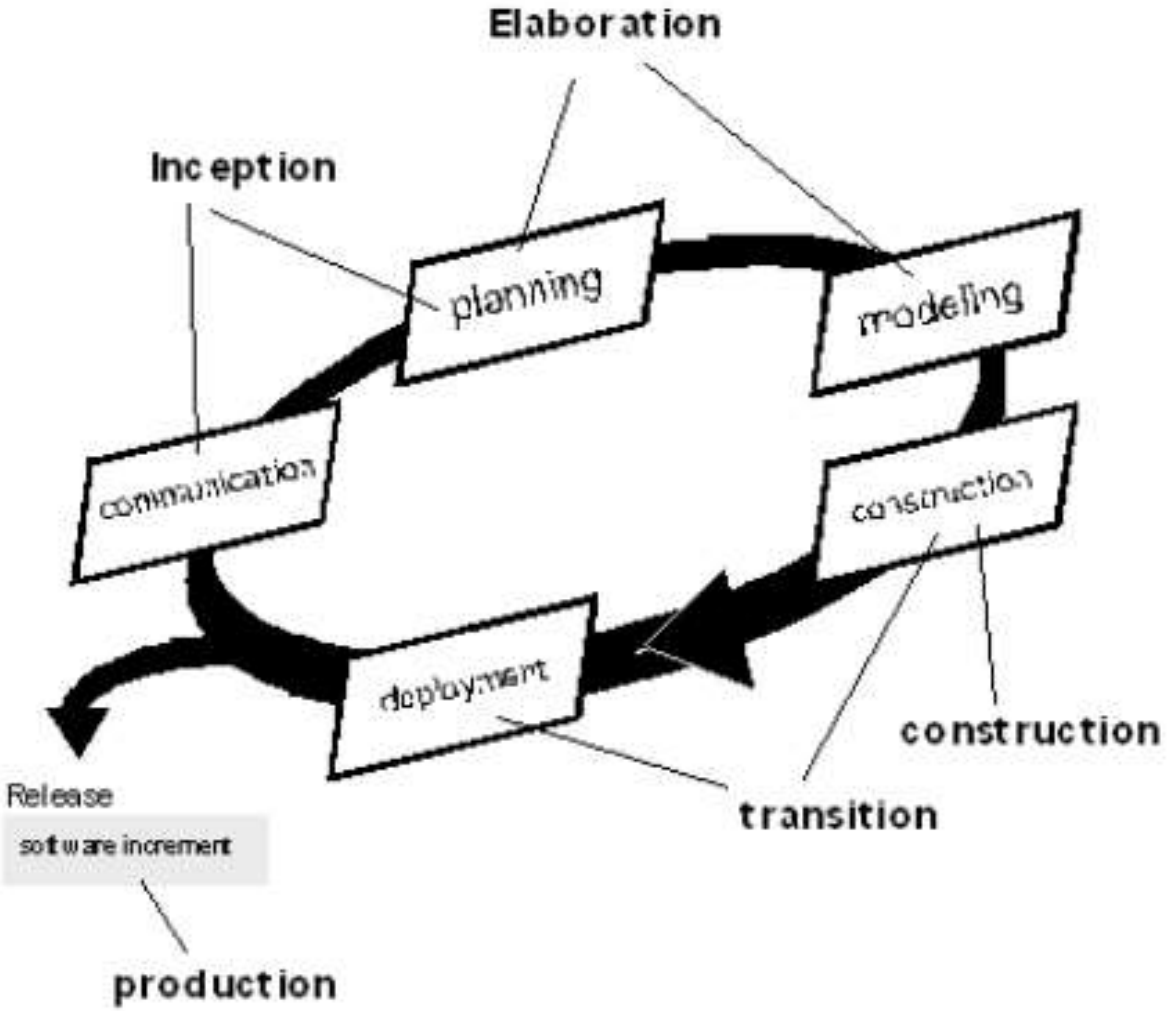
Specialized Process Models

- **Component based development**—the process to apply when reuse is a development objective (like spiral model)
- **Formal methods**—emphasizes the mathematical specification of requirements (easy to discover and eliminate ambiguity, incompleteness and inconsistency)
- **Aspect Oriented software development (AOSD)**—provides a process and methodological approach for defining, specifying, designing, and constructing *aspects*
- **Unified Process**—a "use-case driven, architecture-centric, iterative and incremental" software process closely aligned with the Unified Modeling Language (UML) to model and develop object-oriented system iteratively and incrementally.

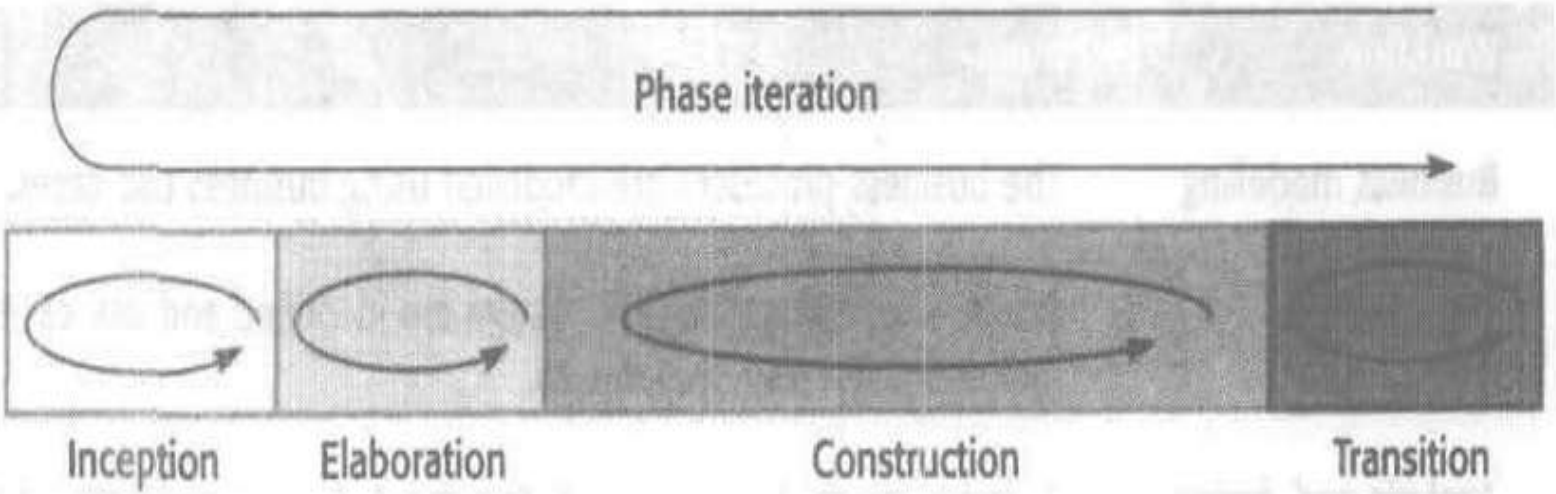
The Unified Process

- The unified process recognize the customer communication and stream lined methods for **describing the customer's view of a system**.

- It emphasizes the importance of **s/w architecture** and helps the architect focus on the right goals


Elaboration
Inception
planning
modeling
communication
construction
deployment
construction
Release
software increment
transition
production

UML provides the necessary technology to support OO s/w engineering practices which does not provide the process frame work to guide project teams in their application of technology



Inception

- Defines the scope of the system
- Identify critical risk
  - The goal of the inception phase is to establish a business case for the system.
  - You should identify all external entities (people and systems) that will interact with the system and define their interactions

The output of inception phase

- The major shareholders agree on the scope of the project
- It addresses high level requirements
- As the business case is produced ,it helps to provide **green signal to continue with the project**

Elaboration:

- The goals of the elaboration phase are to develop an understanding of the problem domain.
- Establish an architectural framework for the system, develop the project plan and identify key project risks.

Construction:

- The construction phase is essentially concerned with system , programming and testing.

- On completion of this phase, you should have a working software system and associated documentation that is ready for delivery to users.

Transition:

- The final phase of the UP is concerned with moving the system from the development community to the user community and making it work in a real environment.
- During this phase the team focuses on correcting defects and modifying the system.
- The goal of this phase to roll out fully functional system to customers

- If ISRO is building a satellite s/w, which model will you choose and why?